

Kapil Kumar

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SUMMARY

AI Engineer with 5+ years of experience in Agentic AI, LLM application development, and classical ML. Skilled in designing and deploying scalable GenAI solutions using LangChain, LangGraph, RAG, and AWS. Experienced in automating enterprise workflows and implementing responsible AI practices aligned with ISO 42001, NIST AI RMF.

EDUCATION

National Institute of Technology Karnataka (NITK) <i>M.Tech in Computational and Data Science</i>	Karnataka, India 2023 – 2025
Indian Institute of Technology (IIT) Roorkee <i>M.Tech in Pulp and Paper Technology</i>	Uttarakhand, India 2018 – 2020
Amity University Rajasthan <i>B.Tech in Chemical Engineering</i>	Rajasthan, India 2013 – 2017

SKILLS

Programming	Python, JavaScript/TypeScript, SQL
AI/LLM Frameworks	LangChain, LangGraph, LlamaIndex, HuggingFace, CrewAI, MCP, RAG Pipelines
GenAI Models	GPT-4o/4.1, Azure OpenAI, Claude Sonnet, Gemini, Groq LLM, Llama 3.1
Voice & Multimodal	Azure Speech Services (STT/TTS), Whisper, OpenAI Realtime API, WebRTC, VAD, SSML
Search/Retrieval	Azure Cognitive Search, FAISS, Pinecone, Qdrant, BM25
Backend APIs	FastAPI, Flask, NestJS, REST APIs, OAuth2.0, JWT, RBAC
Data/Vector Stores	PostgreSQL, MongoDB, MySQL, Redis, Qdrant, Pinecone, FAISS
Frontend	Next.js, React, WebSockets, Tailwind
Cloud/DevOps	Azure (AKS, ACR, App Services, Application Insights), AWS (Lambda, EKS, CloudWatch), Docker, GitHub Actions, CI/CD, AWS Bedrock
Monitoring/Observability	Azure Application Insights, Prometheus, Grafana
Security/Governance	DTAC, NHS DSP Toolkit, ISO 42001, NIST AI RMF, ISO 27001, GDPR
No-code Tools	n8n, Langflow, Cursor
Visualization	Tableau, Matplotlib, Seaborn, Excel
Development Tools	Jupyter Notebook, VS Code, Postman

EXPERIENCE

Security Analyst – AI & LLM Engineering <i>Netradyne Technology India Pvt. Ltd.</i>	Jan. 2025 – Present <i>Bengaluru, Karnataka, India</i>
- Built a full-stack AI Chatbot for RFP Processing using LangChain, GPT-4o, RAG, and Qdrant, reducing manual review by 70% and improving retrieval accuracy (<i>LangChain, GPT-4o, RAG, Qdrant, Flask, FastAPI, PostgreSQL, React, Docker, AWS</i>). - Mapped AI governance controls from ISO 42001, NIST AI RMF, and EU AI Act to enterprise policies, strengthening responsible AI adoption.	

Research Assistant <i>National Institute of Technology Karnataka (NITK)</i>	Aug. 2024 - June 2025 <i>Surathkal, India</i>
- Built an AI-driven retail assistant using Retrieval-Augmented Generation (RAG) and Agentic AI, leveraging AWS Bedrock foundation models to generate personalized, verified product recommendations from catalogs and customer reviews. - Designed and deployed a scalable RAG pipeline using LangChain, FAISS, and AWS Bedrock, enabling secure orchestration of foundation models and improving product relevance by 80%.	

SE - Data Science <i>Valmet Technologies Engineering Services Pvt. Ltd.</i>	Apr. 2022 – Aug. 2023 <i>Gurugram, Haryana, India</i>
- Built a YOLOv8 vision system detecting black-spot defects across six zones, boosting recall by 18% and cutting downtime by 22% via AWS SageMaker and EKS CI/CD. - Collaborated with cross-functional teams to implement data-driven solutions, resulting in a 25% improvement in KPIs and increased stakeholder satisfaction.	

Assistant Manager- Analytics

JK Paper Ltd.

Dec. 2020 – Mar. 2022

Songadh, Gujarat, India

- Developed ML and optimization models to minimize pulp and chemical costs by 8.6 percent while maintaining GSM, brightness, and tensile strength. Analyzed 25K+ production records using Random Forest to identify key quality drivers and visualize process insights in Tableau.
- Built interactive Tableau dashboards to monitor vacuum variation, leading to a 30% reduction in paper breaks and 25% reduction in energy consumption per tonne.

Research Assistant

IIT Roorkee

May 2019 – Jun. 2020

Roorkee, Uttarakhand, India

- Developed an ML framework using Random Forest, XGBoost, Ridge Regression, and Stacking Ensembles to predict paper quality and optimize energy usage across wet-end to dry-end operations, achieving more than 90% accuracy and enabling 6–8% energy savings Python, scikit-learn, PCA, SHAP, Genetic Algorithm.

PROJECTS

AI Chatbot for RFP Processing | *LangChain, GPT-4o, RAG, Qdrant, React, Flask, FastAPI, PostgreSQL, Docker, AWS*

- Engineered a full-stack RFP chatbot leveraging **LangChain** with **Qdrant** + **Ollama** embeddings for document retrieval and **GPT-4o** for refined response generation.
- Developed frontend in **React/Node.js** with **OAuth2.0, JWT, & RBAC** for secure access control; built backend APIs with **Flask/FastAPI** and **PostgreSQL** for Storing chat history. Leveraged Cursor for UI development.
- Automated RFP analysis pipeline, reducing manual review by **70%** and streamlining compliance workflows.

Mini-EMMA: Azure Voice + Chat RAG Receptionist | *Azure OpenAI, Cognitive Search, Speech Services, FastAPI, NestJS, Next.js, NLP*

- Designed and deployed **Mini-EMMA**, an Azure-hosted voice and chat **RAG receptionist** integrating **Azure OpenAI, Cognitive Search, and Speech Services**; developed a full **prompt-engineering framework**, implemented **NHS-aligned DTAC/DSP safety guardrails**, and added system **observability** using **Azure Application Insights**.
- Built distributed microservices using **FastAPI** and **NestJS**, developed the frontend in **Next.js**, and created an end-to-end **CI/CD pipeline** with **GitHub Actions, ACR, and AKS**; additionally implemented **E2E testing**, conducted **prompt A/B evaluation**, and led a structured **clinician validation** process to ensure reliability and safety.

HireFast – Agentic AI HR Automation | *LangChain, LangGraph, Groq LLM, Flask, OAuth2.0, Docker | Git*

- Built an AI-powered recruitment automation platform using **LangChain, LangGraph, and LLM agents** to orchestrate workflows for job description generation, resume parsing, candidate ranking, and interview scheduling.
- Integrated with **LinkedIn, Google Sheets, and Google Calendar** via **OAuth** for seamless end-to-end hiring automation.
- Deployed secure, containerized architecture with microservices, enabling scalability, modularity, & maintainability.

Anomaly Detection in Paper Manufacturing | *YOLOv8, AWS Glue, SageMaker, EKS, CI/CD* Apr 2022 – Aug 2023

- Developed an AI-based system to detect black spots in real-time using **YOLOv8** across six camera zones.
- Automated data labeling and augmentation with **AWS Glue**; managed training via **Amazon SageMaker**.
- Implemented **CI/CD** with **AWS Lambda, CodePipeline, and CloudWatch**; deployed model to EKS with **A/B testing**, improving recall by **18%** and reducing downtime by **22%**.

Raw Material Optimization in Paper Manufacturing | *Python, scikit-learn, Random Forest, Pandas, Tableau*

- Developed and fine-tuned a **Random Forest** regression model to predict and optimize pulp and chemical usage while maintaining key quality metrics such as GSM, brightness, and tensile strength.
- Conducted feature engineering, hyperparameter tuning (GridSearchCV), and cross-validation on **25K+** production records, achieving an **R²** of **0.91**.
- Performed feature analysis to identify critical process parameters and visualized model insights through **Tableau** dashboards for data-driven process optimization.

LEADERSHIP ROLES

- Led 5-member team at JK Paper, improving efficiency and operations.
- Organized large-scale campus events as Hobbies Club Secretary at IIT Roorkee and led 3 cultural programs at Amity University with 500+ participants.

CERTIFICATIONS

- Completed certifications in cybersecurity, AI governance, LLM development, and API/email security across platforms including Google, Securiiti, and Adobe.